

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

**Time Duration: 1 Hour
Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I S. Praveen (192524239) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

204 15 + 21 + 16 + 10
= 85

①

i) Smart device Connectivity:

Let $W \rightarrow$ Smart TV connected to WIFI

$I \rightarrow$ Smart TV can access the internet

Knowledge Base:

$W \rightarrow I$ Goal: I
 $W \quad I$

ii)

$W \rightarrow I$

$P \rightarrow Q = \neg P \vee Q$

$\neg W \vee I$

Fact W
 $\neg I$

the causes are,

$C1: \neg W \vee I$

$C2: W$

$C3: \neg I$

iii) Resolution steps:

$(\neg W \vee I) \cdot W$, cancel W & $\neg W$
 I

Resolve $I, \neg I$

$\therefore \square$

iv)

The smart TV can access the internet

v)

Machine Auto shutdown:

$O \rightarrow S$

O

Goal: S

②

i) CNF conversion

$$O \rightarrow S$$

$$\neg O \vee S$$

$$\text{Fact: } \boxed{O}$$

$$\text{Negated goal: } \boxed{\neg S}$$

clauses

$$C1: \neg O \vee S$$

$$C2: O$$

$$C3: \neg S$$

ii) $(\neg O \vee S), O$

$$\boxed{S}$$

$$S, \neg S$$

$$\therefore \boxed{\square}$$

iv)

The machine shuts down automatically

3) i) Amount Verification:

$$V \rightarrow A$$

$$\text{goal: } A$$

ii)

$$V \rightarrow A$$

$$\neg V \vee A$$

$$\begin{array}{l} V \\ \text{Negated goal:} \\ \neg A \end{array}$$

clauses:

$$C1: \neg V \vee A$$

$$C2: V$$

$$C3: \neg A$$

iii) Resolution steps:

$$(\neg V \vee A), V$$

$$\boxed{A}$$

③

$A, \neg A$

$\therefore \square$

iv)

The user's account is verified

4)

Premium Subscription Access

i)

$P \rightarrow A$

P

Goal: A

ii)

$P \rightarrow A$

$\neg P \rightarrow A$

P

$\neg A$

causes:

$C1: \neg P \vee A$

$C2: P$

$C3: \neg A$

iii)

$(\neg P \vee A), P$

A

$A, \neg A$

$\square$

iv)

Another Gets Premium Subscription

5)

Online Shipping Confirmation:

i)

$P \rightarrow C$

$C \rightarrow S$

P

Goal: S

Q.

ii)

$$P \rightarrow C$$

$$P \rightarrow Q = \neg P \vee Q$$

$$\neg P \vee C$$

$$C \rightarrow S$$

$$\neg C \vee S$$

P

iii)

S

$\neg S$

clauses:

$$C1: \neg P \vee C$$

$$C2: \neg C \vee S$$

$$C3: P$$

$$C4: \neg S$$

iv)

$$C1: \neg P \vee C$$

$$C2: P$$

Resolve



confirmed

$$\neg C \vee S$$

C

$$\boxed{S}$$

$\therefore$

S

$\neg S$

S, $\neg S$

$$\boxed{\quad}$$